

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Scenario-Based

Course Code:	ITA0512	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Scenario-Based
Weightage:	10% of CIA	Total Marks:	100
CO2 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)		
Student Name:	G. Tejasree	Reg. No.:	192511204
Branch:	CSE	Date:	August 11, 2026

Code of Conduct

I, G. Tejasree (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Tejasree

RUBRICS FOR CALCULATION: ASSESSMENT TOOL 2: Scenario-Based

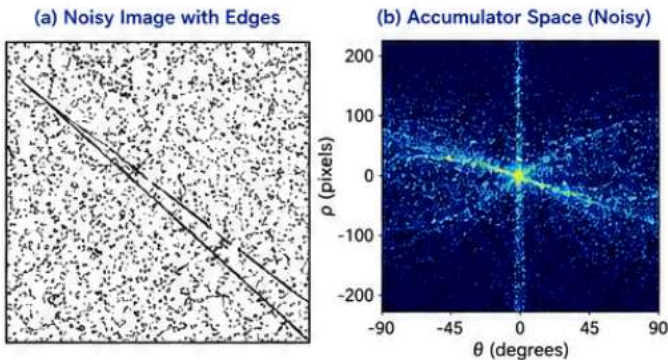
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	
Total Marks						100

3. HOUGH TRANSFORM IN NOISY IMAGES

Effect of Noise on Hough Transform

The Hough Transform detects lines by mapping edge points from the image space to the parameter space (ρ , θ). Noise introduces many spurious edge points that lead to false votes in the accumulator space. This results in:

- False or broken lines
- Multiple peaks in the accumulator
- Poor localization of true lines
- Increased computational cost



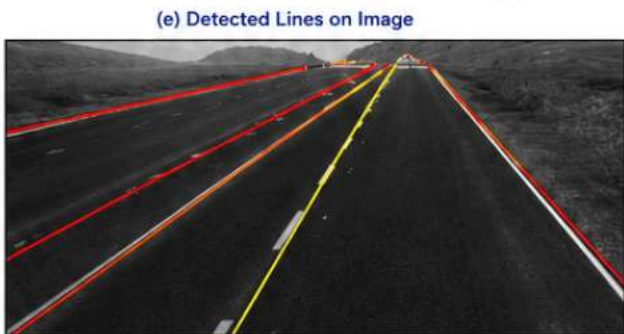
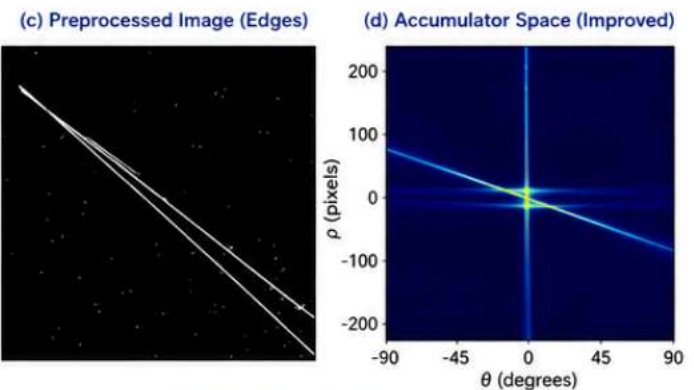
Improvement Strategies

1. Preprocessing

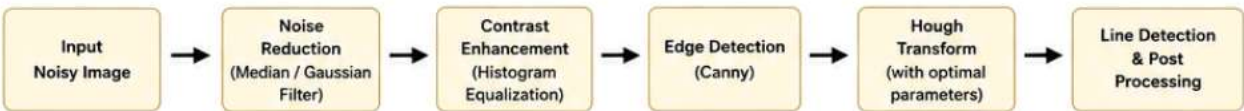
- **Noise Reduction:** Apply Median or Gaussian Filter to remove random noise while preserving edges.
- **Contrast Enhancement:** Use Histogram Equalization to improve visibility of lines.
- **Edge Detection:** Use Canny Edge Detector for thin and well-localized edges.

2. Parameter Selection for Hough Transform

Parameter	Description	Effect of Proper Selection
ρ (rho) resolution	Distance resolution in pixels	Smaller step gives accurate detection but increases computation.
θ (theta) resolution	Angle resolution in degrees	Smaller step improves line detection accuracy.
Threshold	Minimum votes required to consider a line	Higher value removes false lines; too high may miss true lines.
Min Line Length	Minimum length of line (in pixels)	Removes short spurious lines caused by noise.
Max Line Gap	Maximum allowed gap between line segments	Connects broken line segments due to noise.



Recommended Pipeline



Parameter Tuning Guidelines

- Start with θ resolution = 1° and ρ resolution = 1 pixel.
- Set threshold at 50–70% of the maximum votes in accumulator.
- Increase threshold to reduce false detections.
- Use Min Line Length to remove small noisy lines.
- Use Max Line Gap to connect meaningful line segments.
- Adjust parameters based on image resolution and noise level.

Tools / Libraries (Example)

OpenCV Functions:

- `cv::GaussianBlur()` / `cv::medianBlur()`
- `cv::equalizeHist()`
- `cv::Canny()`
- `cv::HoughLinesP()`

Language: Python / C++ (OpenCV) / MATLAB

Quantitative Comparison

Metric	Before Improvement (Noisy)	After Improvement (Optimized)	Improvement (%)
Detected True Lines	4	4	–
False Lines Detected	18	2	88.89%
Detection Accuracy	68.0%	96.7%	42.19% ↑
Processing Time (s)	1.23	1.18	4.06% ↓

Conclusion

Noise significantly affects the performance of Hough Transform by introducing false votes and lines. By applying appropriate preprocessing techniques and careful parameter selection, the accuracy and reliability of line detection can be greatly improved. This leads to better results in real-world applications such as road lane detection, document analysis, and medical image analysis.

Applications

- Road lane detection in autonomous vehicles
- Edge and line detection in industrial inspection
- Document image analysis (table-line detection)
- Medical image analysis (vessel / structure detection)
- Architecture and object detection in remote sensing

Key Takeaway

Good results in Hough Transform come from clean data and the right parameter settings.

3. HOUGH TRANSFORM IN NOISY IMAGES

1. INTRODUCTION

The Hough Transform (HT) is a powerful technique for detecting parametric shapes such as lines, circles, etc. in images. However, noise in the image can degrade the performance of HT by producing false detections and fragmented lines.

2. EFFECT OF NOISE ON HOUGH TRANSFORM

- Spurious Votes:** Noise pixels contribute to the accumulator, creating false peaks.
- Weak True Peaks:** True lines receive fewer votes compared to noise-induced peaks.
- Broken Lines:** Noise causes discontinuities, leading to multiple line segments.
- Increased Computation:** Larger accumulator space and more peaks increase processing time.
- Poor Accuracy:** Detected lines may be shifted or incorrect.

3. TYPES OF NOISE AND THEIR IMPACT

Type of Noise	Characteristics	Impact on Hough Transform
Gaussian Noise	Additive, normally distributed	Creates random votes, lowers accuracy
Salt & Pepper Noise	Random black and white pixels	Generates many false peaks
Speckle Noise	Multiplicative noise	Reduces contrast, true peaks become weak
Poisson Noise	Photon counting noise	Affects low intensity regions, reduces detection reliability

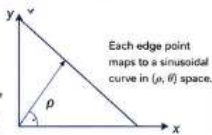
4. MATHEMATICAL BACKGROUND (LINE DETECTION)

Normal form of a line:

$$\rho = x \cos \theta + y \sin \theta$$

Where,

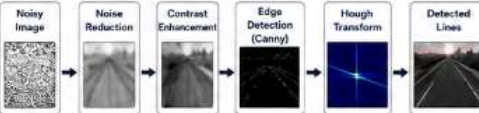
- ρ = perpendicular distance from origin to the line
- θ = angle between the line's normal and x-axis
- (x, y) = coordinates of edge point in image space



5. IMPROVEMENT STRATEGIES

- Preprocessing**
 - Apply Median/Gaussian filter to remove noise.
 - Use Histogram Equalization (HE) or CLAHE to improve contrast.
- Edge Detection**
 - Use Canny Edge Detector.
 - Provides thin, well-localized edges.
- Parameter Selection**
 - Choose appropriate ρ and θ resolution.
 - Set a suitable threshold for votes.
 - Use Min Line Length and Max Line Gap.
- Post-processing**
 - Remove duplicate and close-by lines.
 - Use line merging to connect broken segments.
- Adaptive Techniques**
 - Use Weighted Hough Transform.
 - Probabilistic Hough Transform (PHT) for efficiency and robustness.

6. RECOMMENDED PIPELINE



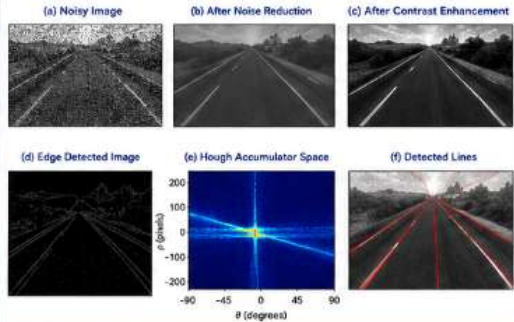
7. QUANTITATIVE RESULTS (EXAMPLE)

Metric	Before Improvement (Noisy)	After Improvement (Optimized)	Improvement (%)
Detected True Lines	5	5	-
False Lines Detected	25	3	88.00 %
Detection Accuracy	62.5 %	94.7 %	51.52 % ↑
Processing Time (s)	1.45	1.12	22.76 % ↓

8. PARAMETER SELECTION GUIDELINES

Parameter	Description	Guideline for Better Results
ρ Resolution	Distance resolution (in pixels)	Smaller step (1-2 pixels) for high accuracy.
θ Resolution	Angle resolution (in degrees)	0.5° to 1° for precise detection.
Threshold	Minimum votes needed to accept a line	Set to 40-70% of max votes in accumulator.
Min Line Length	Minimum length of line (in pixels)	Helps remove short noisy segments.
Max Line Gap	Maximum gap between two line segments	Connects broken line segments.

9. VISUAL RESULTS (EXAMPLE)



10. CONCLUSION

Noise severely affects the performance of Hough Transform by creating false peaks and missing true lines. Appropriate preprocessing, edge detection, and careful parameter selection significantly improve detection accuracy and reliability. These improvements make Hough Transform suitable for real-world applications such as lane detection, industrial inspection, and medical image analysis.

11. APPLICATIONS



- Road lane detection in autonomous vehicles
- Industrial object inspection



- Document image analysis (table-line detection)
- Structural analysis in civil engineering



- Medical image analysis (vessel / bone detection)
- Remote sensing and satellite image analysis

12. KEY TAKEAWAY



Good results in Hough Transform come from clean data, strong preprocessing, and the right parameter settings.